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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CHEMISTRY **5070/11**

Paper 1 Multiple Choice **October/November 2012**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

2931853253

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

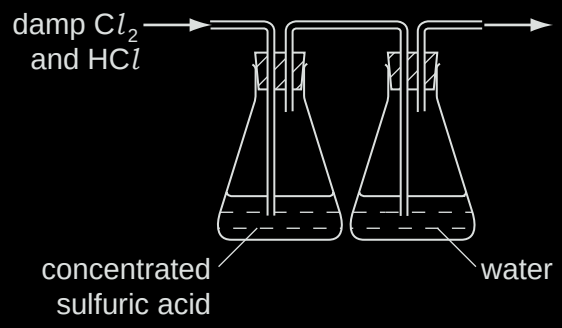
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
A copy of the Periodic Table is printed on page 16.

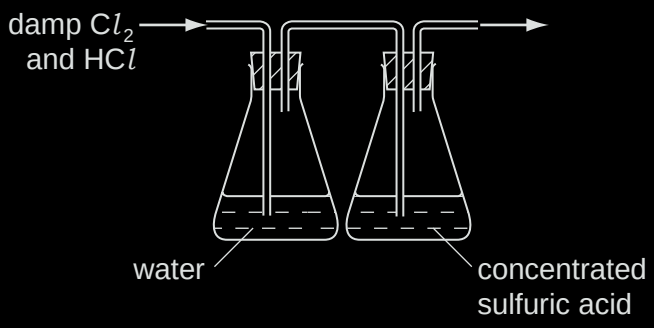
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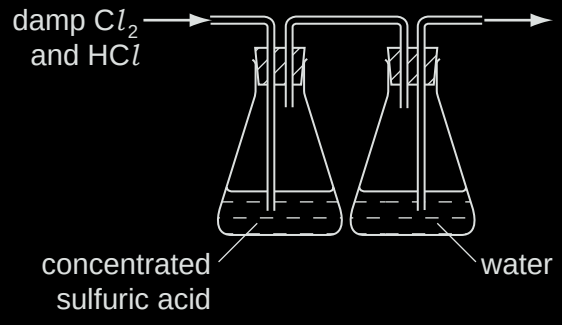
A



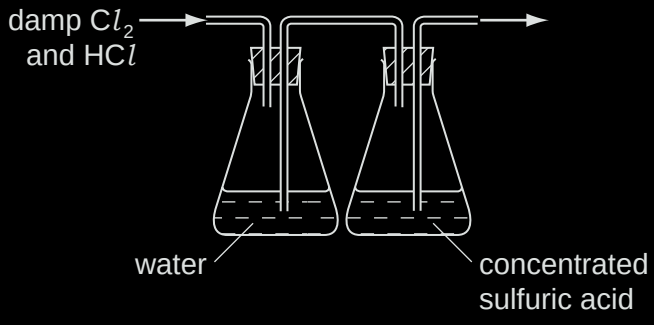
B



C



D



6 Which of the following is **not** a mixture?

A ethanol

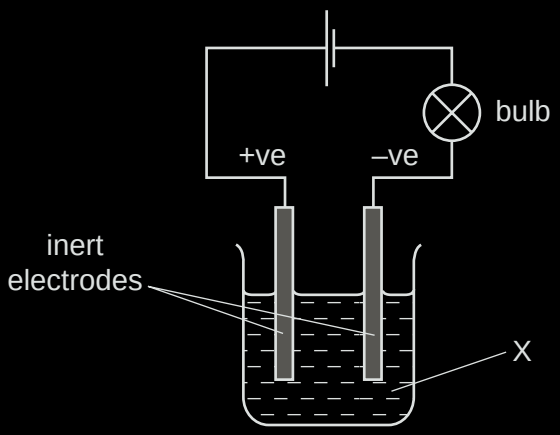
B petrol

C steel

D tap water

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What is X?

A aqueous copper(II) sulfate

B concentrated aqueous sodium chloride

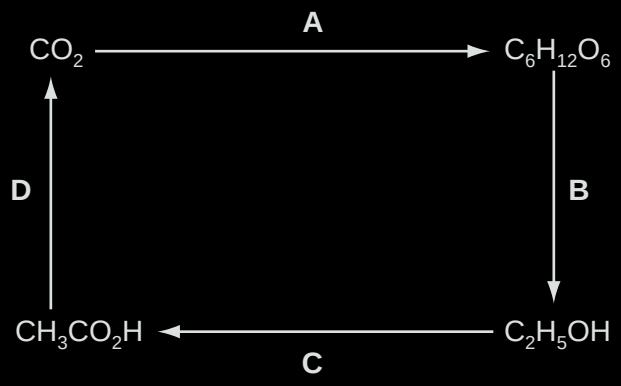
C ethanol

D molten lead bromide

14 Which element in the table is an alkali metal?

	melting point °C	density g / cm^3
A	–39	13.60
B	–7	3.10
C	98	0.97
D	1083	8.92

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16 Which industrial reaction does **not** involve a catalyst?

A the cracking of hydrocarbons

B the extraction of iron from haematite in a blast furnace

C the production of ammonia from nitrogen and hydrogen

D the redox reaction involving the removal of combustion pollutants from car exhausts

17 Salts containing which of the following anions are always soluble in water?

A carbonates

B chlorides

C nitrates

D sulfates

18 What is a property of the hydroxide, OH^-, ion?

A It combines with hydrogen to form water.

B It is present in water.

C It readily breaks down into hydrogen ions and oxide ions.

D It travels to the cathode in electrolysis of an aqueous solution.

19 Which method of preparation of magnesium sulfate is an example of a redox reaction?

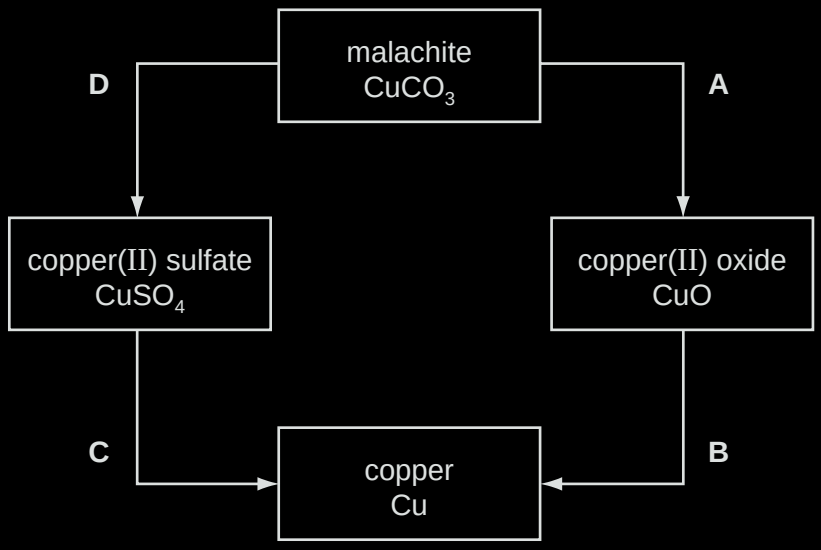
A $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$

B $MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$

C $Mg(OH)_2 + H_2SO_4 \rightarrow MgSO_4 + 2H_2O$

D $MgCO_3 + H_2SO_4 \rightarrow MgSO_4 + H_2O + CO_2$

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21 Which process is a renewable energy source?

A combustion of coal

B electrolysis of aluminium oxide

C fractional distillation of petroleum

D photosynthesis

22 An element X forms an ion X^{3-}.

In which group of the Periodic Table will this element be found?

A Group I

B Group III

C Group V

D Group VII

23 Which two gases do not damage limestone buildings?

A nitrogen and carbon monoxide

B nitrogen dioxide and carbon monoxide

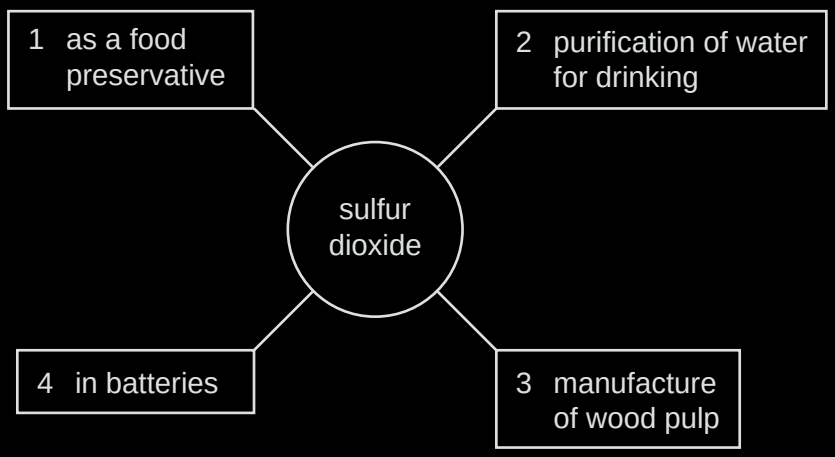
C nitrogen dioxide and carbon dioxide

D sulfur dioxide and carbon dioxide

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Which two of the numbered boxes are correct?

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 and 4

33 Which statement about macromolecules is correct?

A Nylon and *Terylene* are both polyesters.

B Proteins and nylon have the same monomer units.

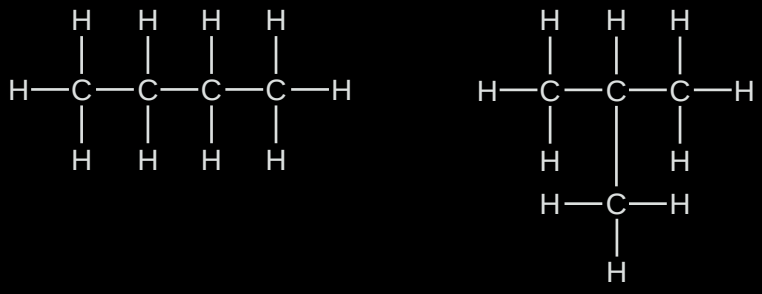
C Proteins have the same amide linkages as nylon.

D *Terylene* and fats are esters but with different linkages.

34 Which row shows both the correct source and the correct effect of the named pollutant?

	pollutant	source	effect
A	carbon monoxide	incomplete combustion of carbon-containing materials	global warming
B	oxides of nitrogen	decaying vegetable matter	global warming
C	ozone	photochemical reactions	acid rain
D	sulfur dioxide	volcanoes	acid rain

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It can be predicted from their formulae that the compounds have the same

A boiling point.

B composition by mass.

C melting point.

D structural formula.

36 Which statement concerning isomers is true?

A Diamond and graphite are isomers of each other.

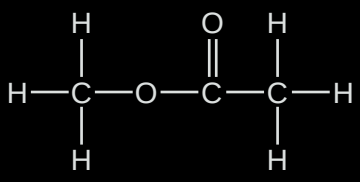
B Isomers have the general formula C_nH_{2n+2}.

C Isomers have the same molecular formula.

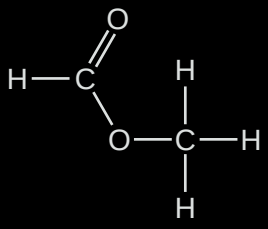
D Macromolecules are isomers of the small molecules from which they are made.

37 Which compound will react with ethanol to form an ester?

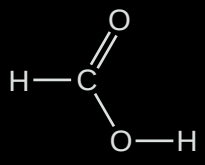
A



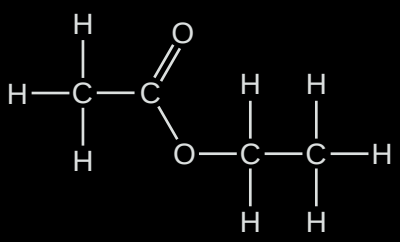
B



C



D



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DATA SHEET
The Periodic Table of the Elements

Group																
I	II										III	IV	V	VI	VII	0
							1 H Hydrogen 1									4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radius 88	227 Ac Actinium 89 †														

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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